

CHAPTER 3

SCALAR

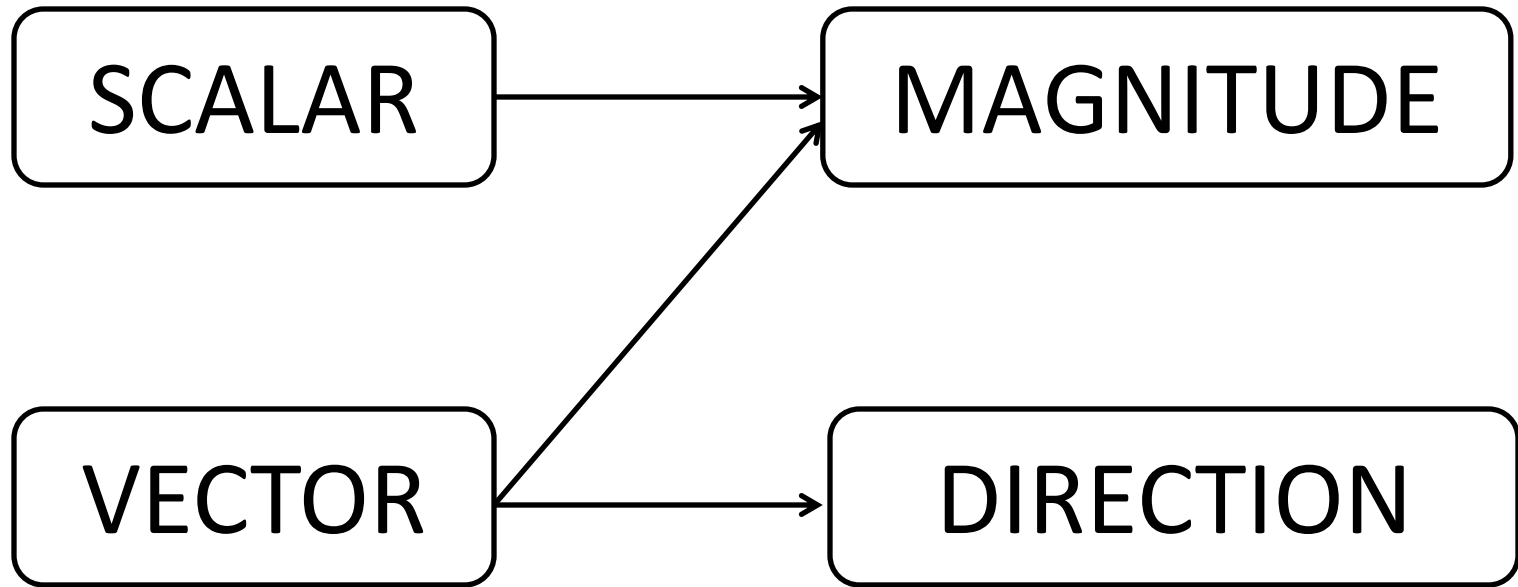
&

VECTOR

LEARNING OUTCOMES

- Express vector
- Operations of vectors
- Scalar (dot) product
- Vector (cross) product

EXPRESS VECTORS



EXPRESS VECTORS

SCALAR

Mass

Distance

Temperature

Current

Power

VECTOR

Weight

Displacement

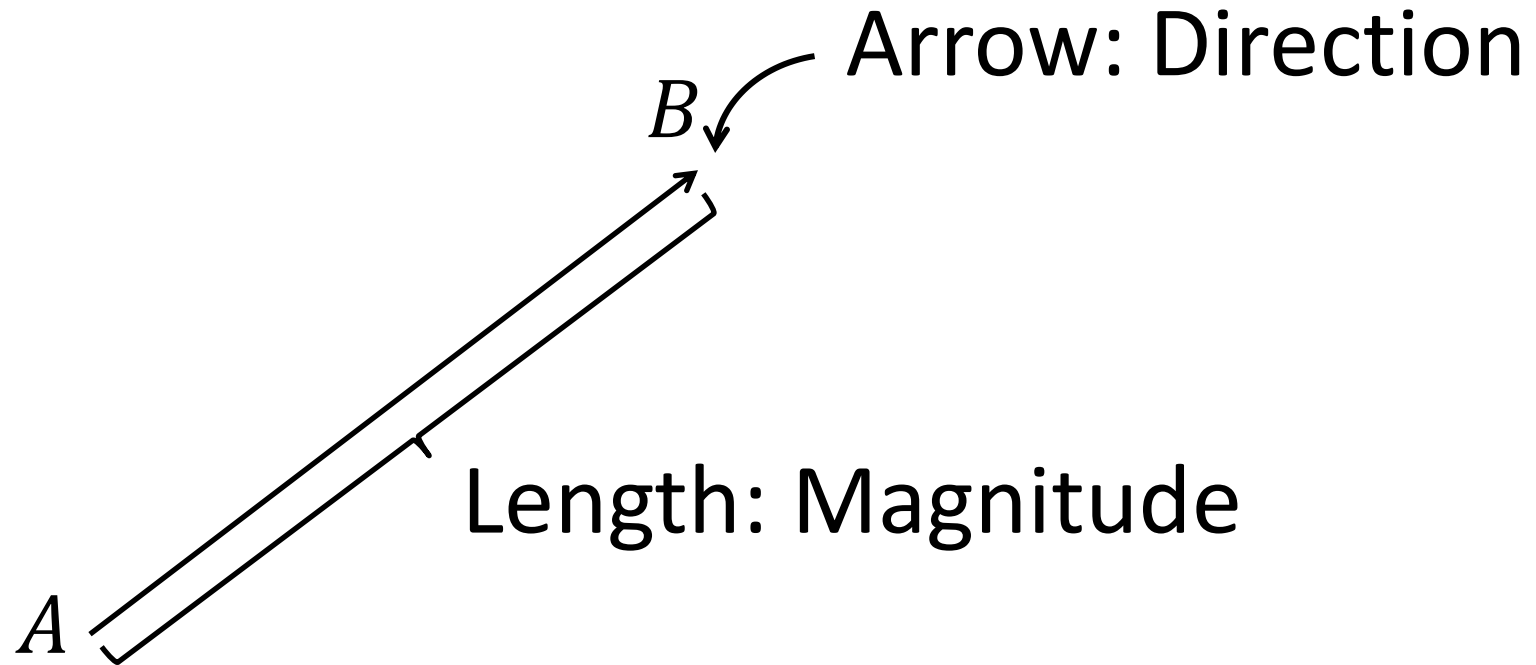
Impulse

Pressure

Moment

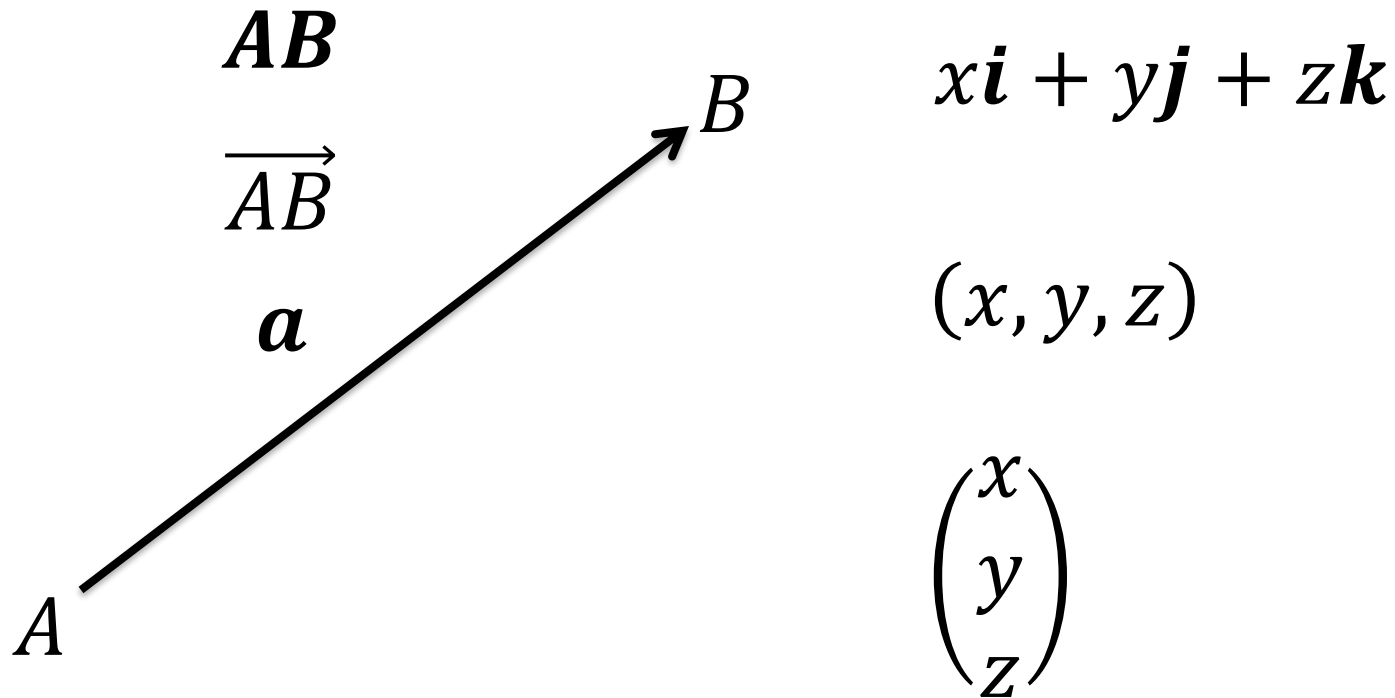
EXPRESS VECTORS

➤ Vector notation & representation



EXPRESS VECTORS

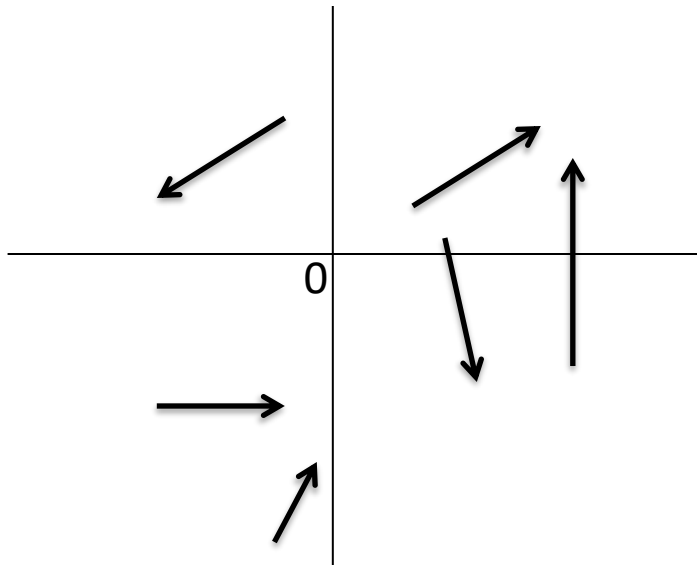
➤ Vector notation & representation



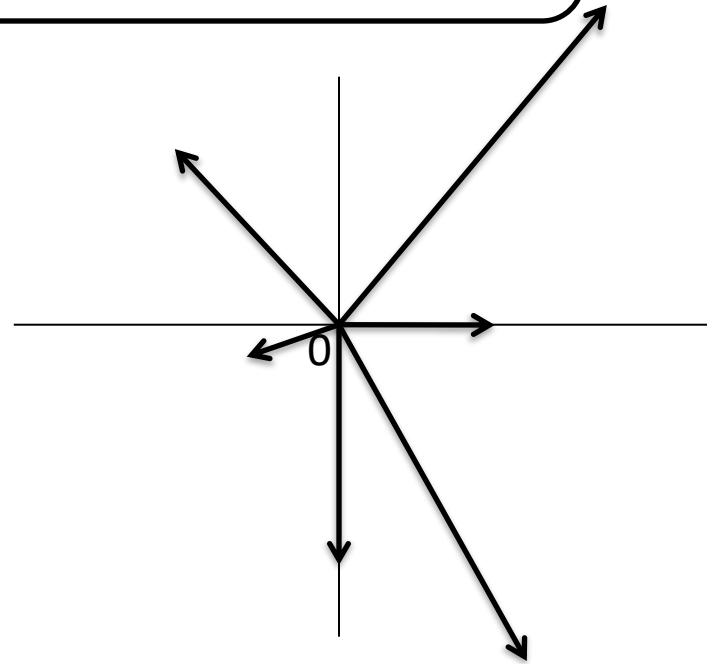
EXPRESS VECTORS

➤ Vector notation & representation

Independent
Vector



Position
Vector



EXERCISE 1

The cost of a theater ticket

SCALAR

The current in a river

VECTOR

The initial flight path from
KL to Amsterdam

VECTOR

EXERCISE 1

The population of the world

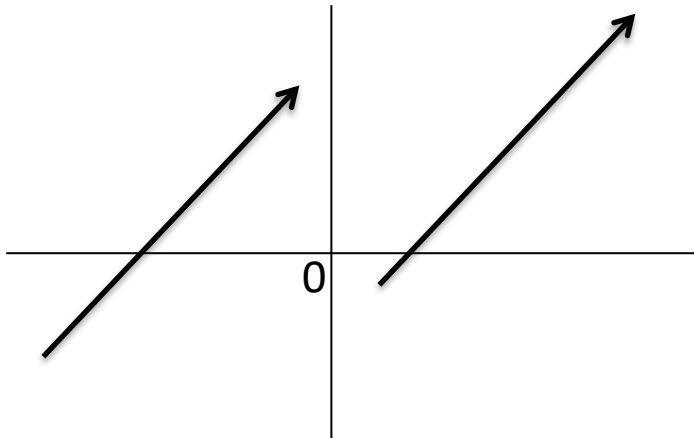
SCALAR

Speed of light

SCALAR

EXPRESS VECTORS

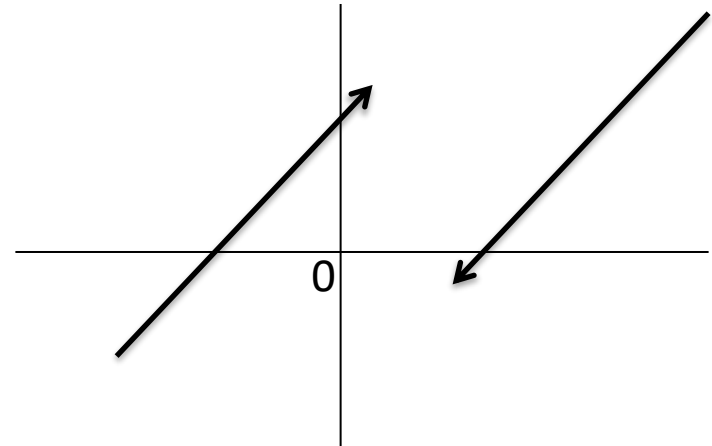
Equality of
Vectors



= length

= direction

Negative
Vector



= length

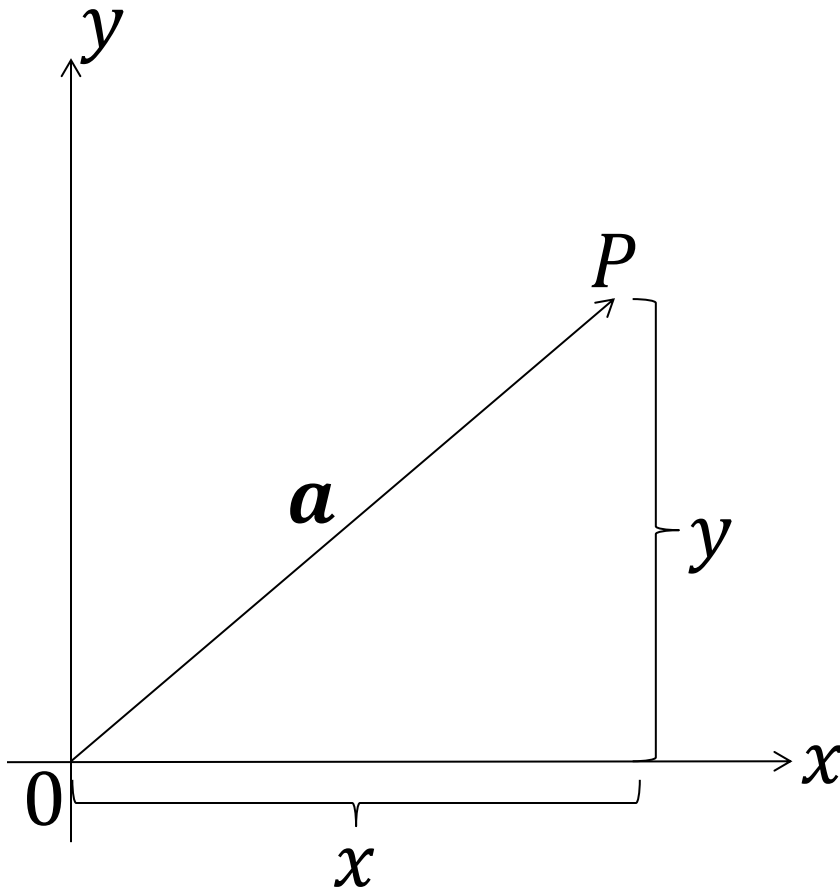
OPPOSITE direction

EXPRESS VECTORS

➤ Calculate the basic vector

MAGNITUDE VECTOR

Length



$$\text{If } \vec{OP} = x\mathbf{i} + y\mathbf{j}$$

$$\therefore |\vec{OP}| = \sqrt{x^2 + y^2}$$

$$\text{If } \vec{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$\therefore |\vec{OP}| = \sqrt{x^2 + y^2 + z^2}$$

EXPRESS VECTORS

➤ Calculate the basic vector

UNIT VECTOR

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

EXERCISE 2

① Given $\vec{p} = 14\mathbf{i} + 11\mathbf{j} - 6\mathbf{k}$.

Find its magnitude and unit vector.

Answer:

Magnitude of $\vec{p}$:

$$|\vec{p}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{14^2 + 11^2 + (-6)^2}$$

$$= \sqrt{353} = 18.788$$

EXERCISE 2

Answer:

Unit vector of $\vec{p}$: $\hat{p} = \frac{\vec{p}}{|\vec{p}|}$

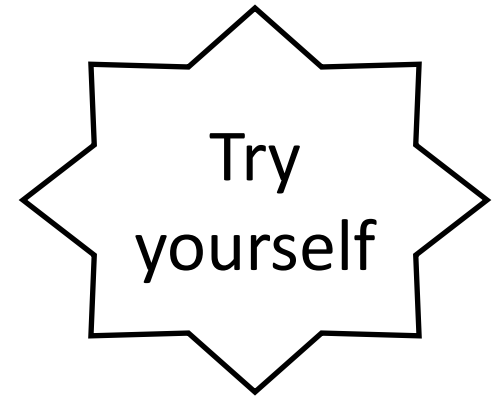
$$\hat{p} = \frac{14\mathbf{i} + 11\mathbf{j} - 6\mathbf{k}}{18.788}$$

$$= \frac{14}{18.788}\mathbf{i} + \frac{11}{18.788}\mathbf{j} - \frac{6}{18.788}\mathbf{k}$$

$$= 0.745\mathbf{i} + 0.585\mathbf{j} - 0.319\mathbf{k}$$

EXERCISE 2

② If $\mathbf{a} : \mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$, find:



a) $\hat{\mathbf{a}}$

b) the magnitude of a vector that is opposite direction of vector $\mathbf{a}$

Answer:

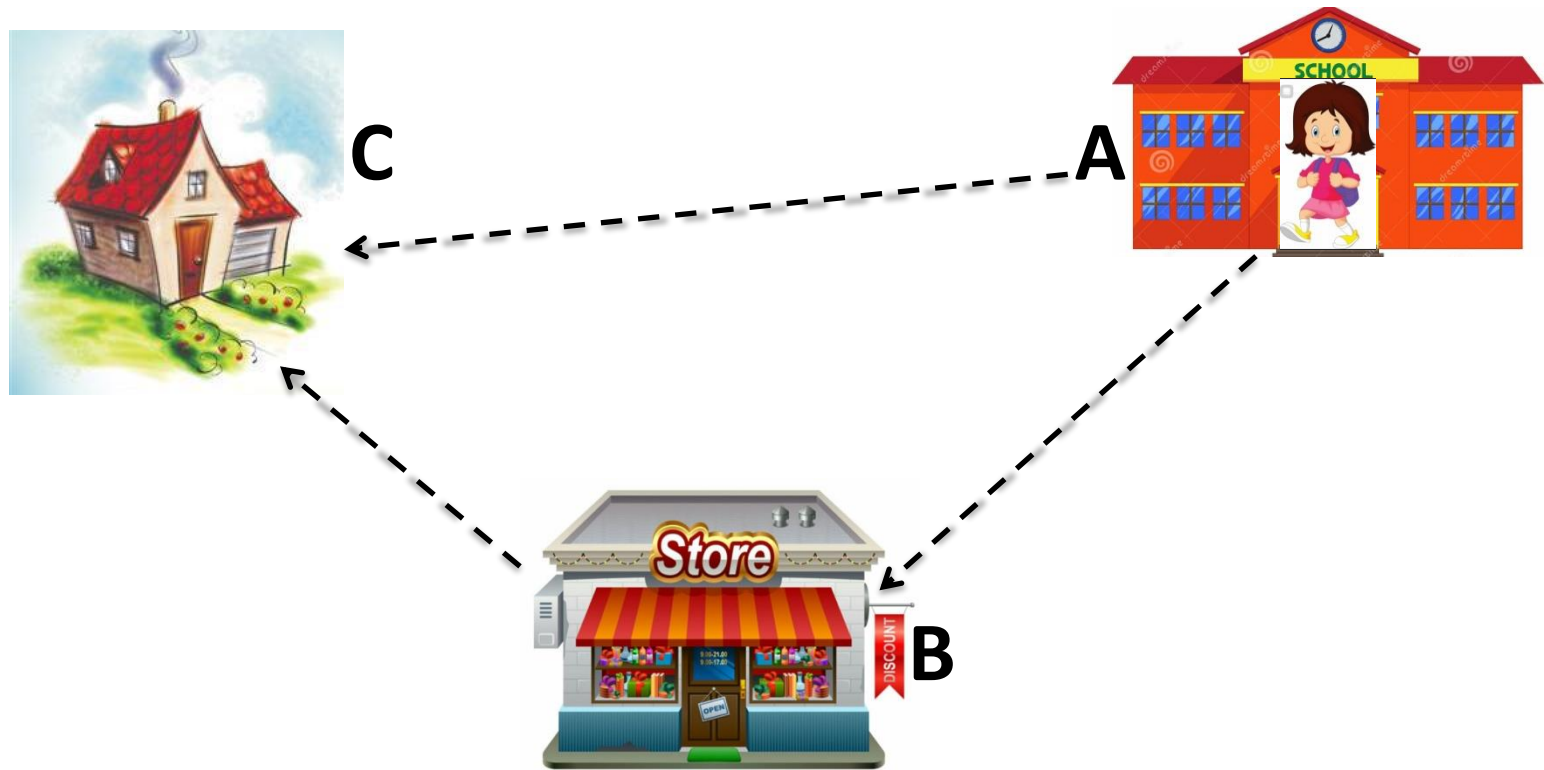
a) $\hat{\mathbf{a}} = 0.169\mathbf{i} - 0.507\mathbf{j} + 0.845\mathbf{k}$

b) $|\mathbf{-a}| = 5.916$

LEARNING OUTCOMES

- Express vector 
- Operations of vectors
- Scalar (dot) product
- Vector (cross) product

OPERATION OF VECTORS

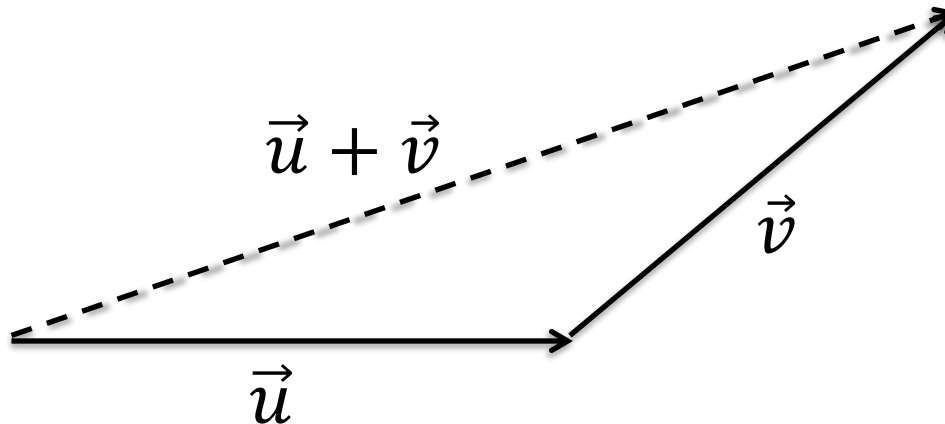


$$\vec{AB} + \vec{BC} = \vec{AC}$$

OPERATION OF VECTORS

Definition: +

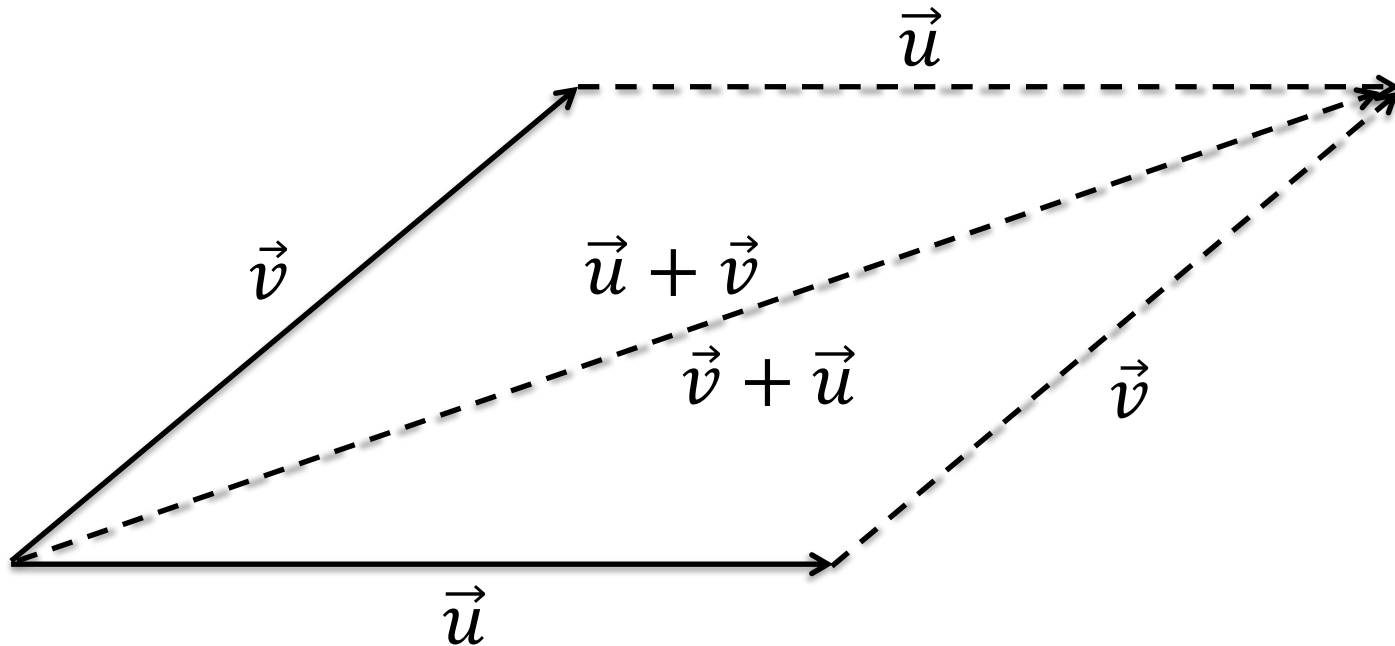
If $\vec{u}$ and $\vec{v}$ are vector positioned so the initial point of $\vec{v}$ is at the terminal point of $\vec{u}$, then the sum $\vec{u} + \vec{v}$ is the vector from the initial point of $\vec{u}$ to the terminal point of $\vec{v}$.



TRIANGLE LAW

OPERATION OF VECTORS

Definition: +



PARALLELOGRAM LAW

OPERATION OF VECTORS

Definition: 

The sum of vector $\vec{u}$ and the negative vector $\vec{v}$

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$$

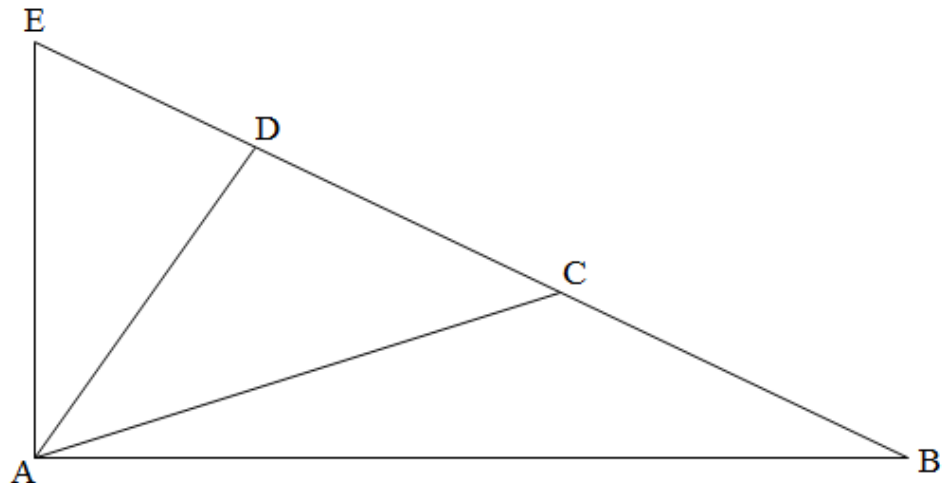
EXERCISE 3

- ① Figure below shows a triangle, ABCDE with a straight line $\overrightarrow{BC} = \overrightarrow{CD} = \overrightarrow{DE}$. Given that, $\overrightarrow{AB} = 3\mathbf{a} + 2\mathbf{b}$ and $\overrightarrow{BC} = \mathbf{a} - \mathbf{b}$. Find the following vectors in the term of $\mathbf{a}$ and $\mathbf{b}$.

a) $\overrightarrow{ED}$

b) $\overrightarrow{AC}$

c) $\overrightarrow{DA}$



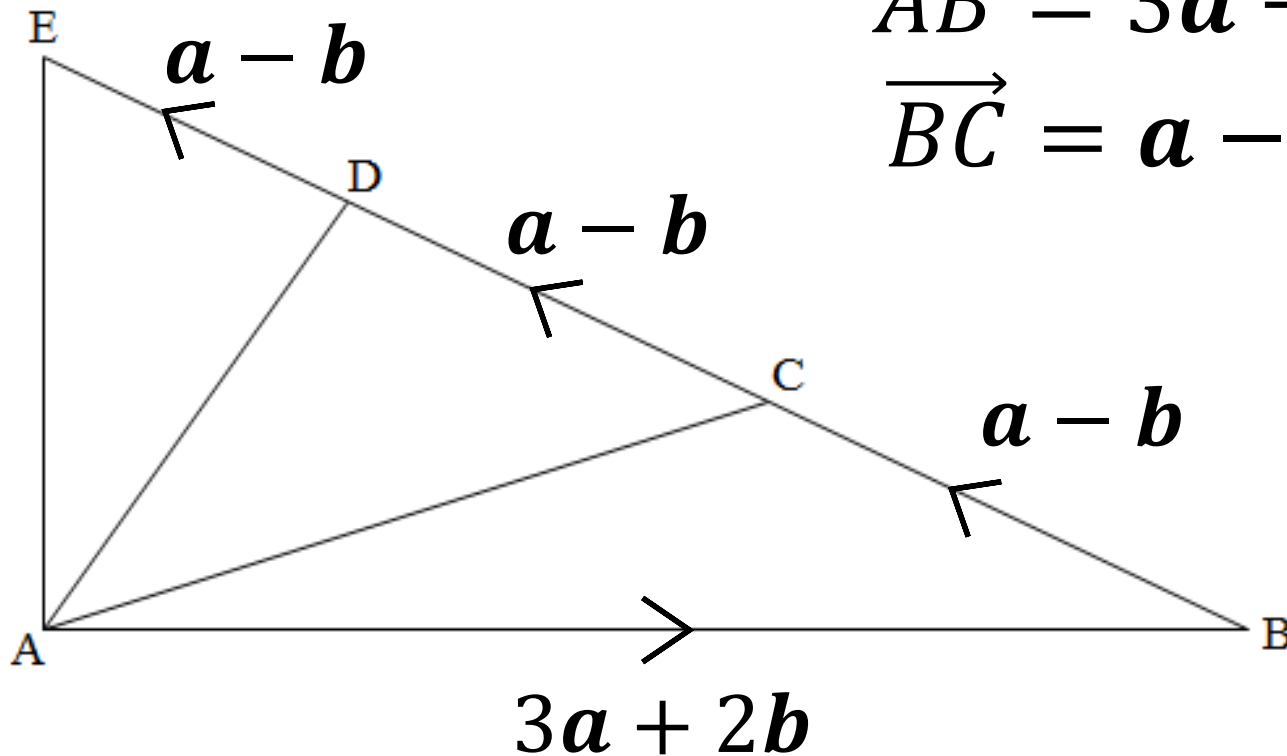
EXERCISE 3

Answer:

Given: $\overrightarrow{BC} = \overrightarrow{CD} = \overrightarrow{DE}$

$$\overrightarrow{AB} = 3\mathbf{a} + 2\mathbf{b}$$

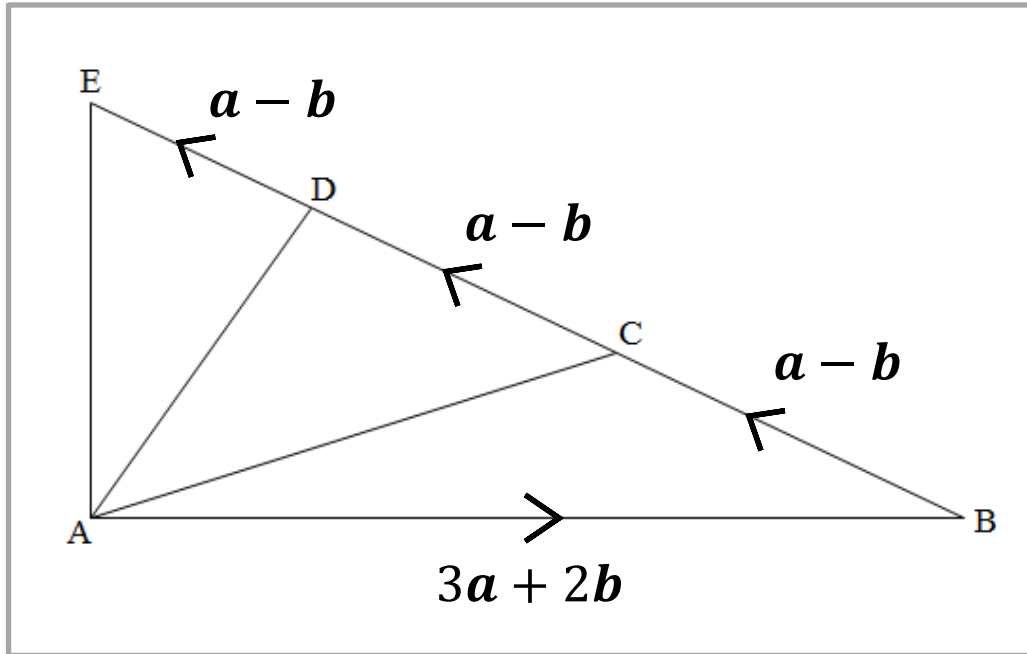
$$\overrightarrow{BC} = \mathbf{a} - \mathbf{b}$$



EXERCISE 3

Answer:

a) $\overrightarrow{ED}$

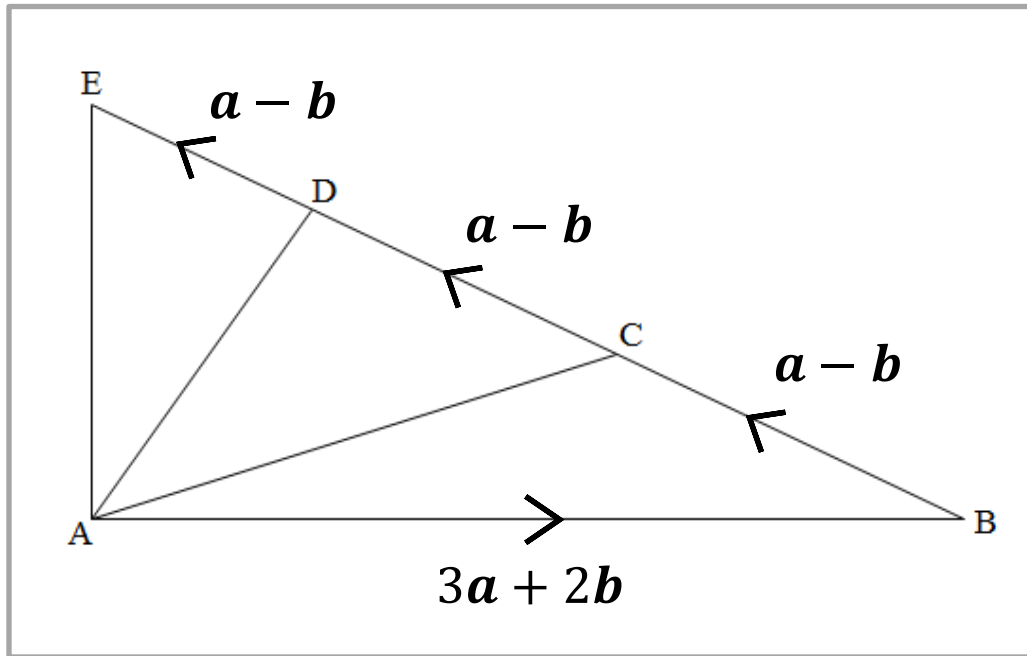


$$\begin{aligned}\overrightarrow{ED} &= -\overrightarrow{DE} \\ &= -(a - b) \\ &= -a + b\end{aligned}$$

EXERCISE 3

Answer:

b) $\overrightarrow{AC}$

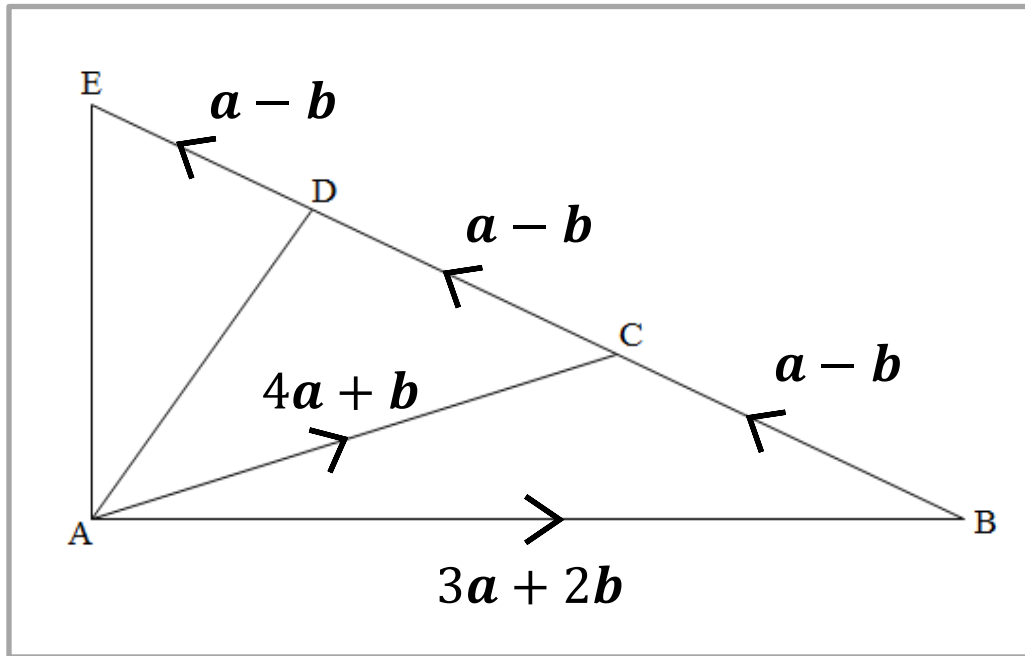


$$\begin{aligned}\overrightarrow{AC} &= \overrightarrow{AB} + \overrightarrow{BC} \\ &= 3a + 2b + a - b \\ &= 4a + b\end{aligned}$$

EXERCISE 3

Answer:

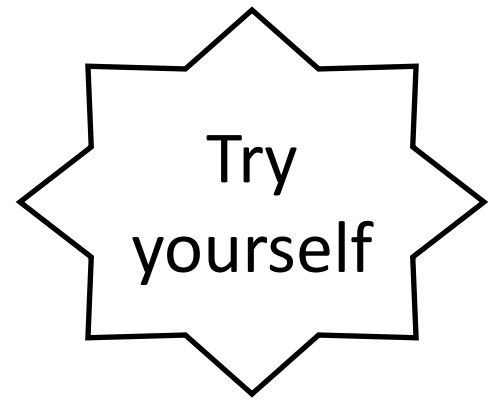
c) $\overrightarrow{DA}$



$$\begin{aligned}\overrightarrow{DA} &= \overrightarrow{DC} + \overrightarrow{CA} \\ &= -a + b + (-4a - b) \\ &= -a + b - 4a - b \\ &= -5a\end{aligned}$$

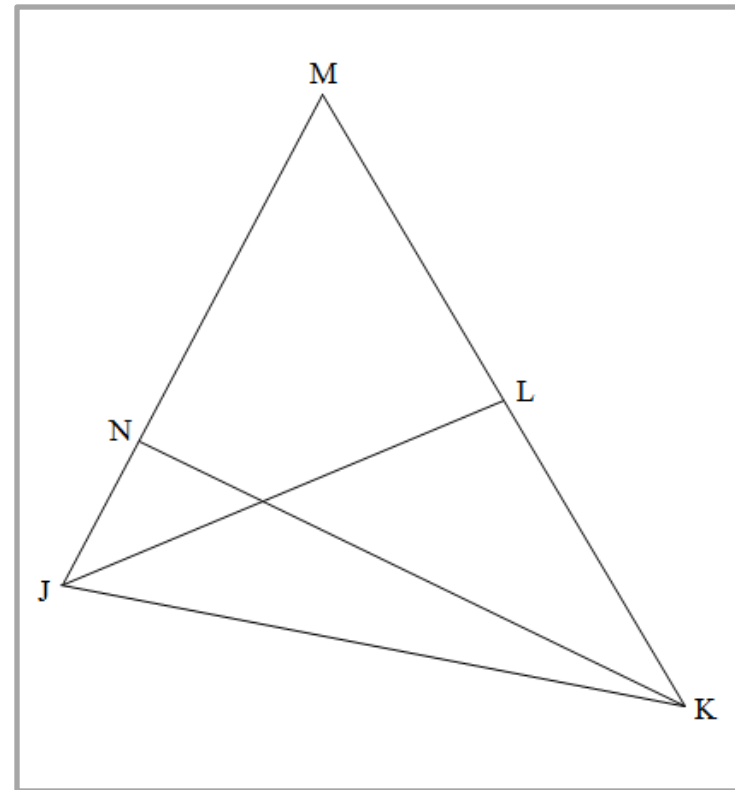
②

EXERCISE 3

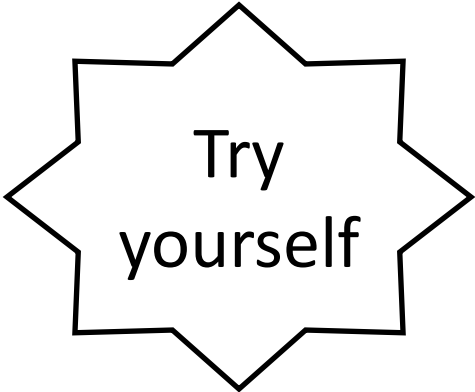


In the diagram, JNM and KLM are straight lines. Points L and N lie on KM and JM respectively such that L is the midpoint of KM and $3JN = JM$. Given $\overrightarrow{JK} = 4\mathbf{a}$ and $\overrightarrow{JM} = 6\mathbf{b}$. Express each of the following vectors in the term of $\mathbf{a}$ and $\mathbf{b}$.

- a) $\overrightarrow{KM}$ b) $\overrightarrow{ML}$ c) $\overrightarrow{JL}$



EXERCISE 3



Try
yourself

② Answer:

a) $-4a + 6b$

b) $2a - 3b$

c) $2a + 3b$

EXERCISE 3

③ If $\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and

$\overrightarrow{ON} = 3\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$, find:

a) $\overrightarrow{MN}$

b) Magnitude of $\overrightarrow{MN}$

c) Unit vector of $\overrightarrow{MN}$

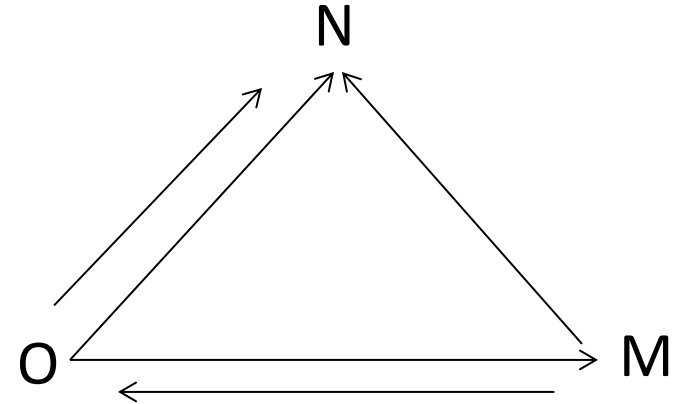
EXERCISE 3

③ Answer:

a) Given:

$$\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$$

$$\overrightarrow{ON} = 3\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$$



$$\overrightarrow{MN} = \overrightarrow{MO} + \overrightarrow{ON}$$

$$= -(4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) + 3\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$$

$$= -4\mathbf{i} + 3\mathbf{j} - 2\mathbf{k} + 3\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$$

$$= -\mathbf{i} + 9\mathbf{j}$$

EXERCISE 3

③ Answer:

$$\text{b) } \overrightarrow{MN} = -\mathbf{i} + 9\mathbf{j}$$

Magnitude of $\overrightarrow{MN}$

$$\begin{aligned} |\overrightarrow{MN}| &= \sqrt{x^2 + y^2 + z^2} \\ &= \sqrt{(-1)^2 + 9^2 + 0^2} \\ &= \sqrt{82} \\ &= 9.055 \end{aligned}$$

EXERCISE 3

③ Answer:

$$\text{b) } \overrightarrow{MN} = -\mathbf{i} + 9\mathbf{j} \quad , \quad |\overrightarrow{MN}| = 9.055$$

Unit vector of $\overrightarrow{MN}$

$$\begin{aligned} \widehat{\overrightarrow{MN}} &= \frac{\overrightarrow{MN}}{|\overrightarrow{MN}|} = \frac{-\mathbf{i} + 9\mathbf{j}}{9.055} \\ &= \frac{-1}{9.055} \mathbf{i} + \frac{9}{9.055} \mathbf{j} \\ &= -0.1110\mathbf{i} + 0.994\mathbf{j} \end{aligned}$$

EXERCISE 3

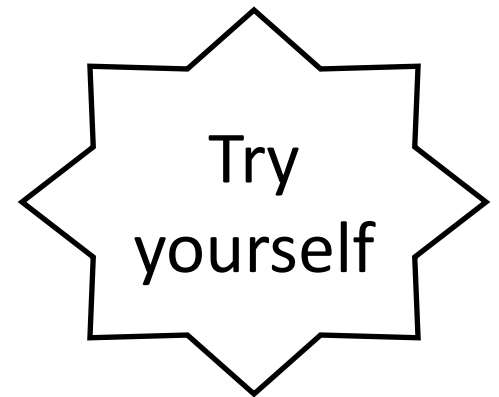
- ④ The position vector of points A and B are $3\mathbf{i} - 5\mathbf{j} + 4\mathbf{k}$ and $2\mathbf{i} + 5\mathbf{j} - \mathbf{k}$ respectively.

Find:

a) $\overrightarrow{AB}$

b) Magnitude of $\overrightarrow{AB}$

c) Unit vector of $\overrightarrow{AB}$



EXERCISE 3

④ Answer:

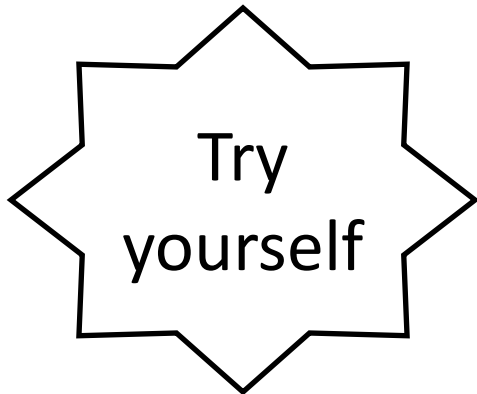
a) $-i + 10j - 5k$

b) 11.225

c) $-0.089i + 0.891j - 0.445k$

EXERCISE 3

⑤ The position vector of P is $7\mathbf{i} + b\mathbf{j} + 2\mathbf{k}$ and the position vector of Q is $4\mathbf{i} - 3\mathbf{j} + 9\mathbf{k}$ where b is a constant. If $\overrightarrow{PQ} = -3\mathbf{i} + 7\mathbf{k}$, what is the value of b ? Hence, find $|\overrightarrow{PQ}|$.



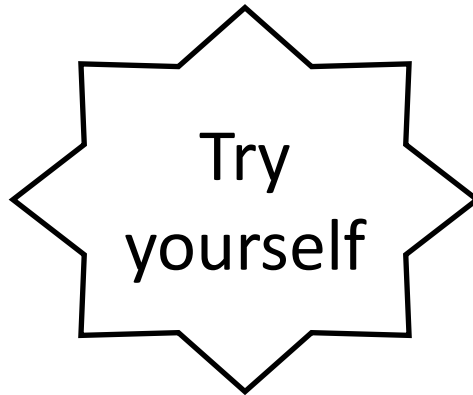
Answer:

$$b = -3$$

$$|\overrightarrow{PQ}| = 7.616$$

EXERCISE 4

- ① Given that $\overrightarrow{OA} = \mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{OB} = 5\mathbf{i} - 7\mathbf{j}$.
Find vector $\overrightarrow{AB}$.

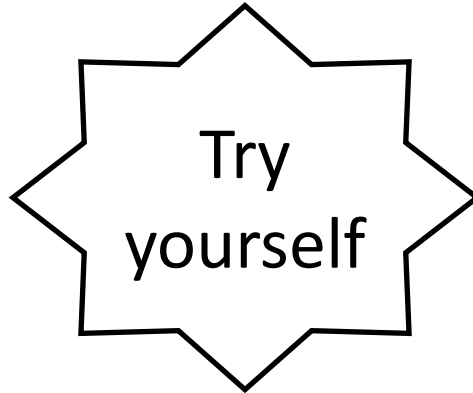


Answer:

$$4\mathbf{i} - 10\mathbf{j}$$

EXERCISE 4

- ② Given that $\overrightarrow{OS} = -5\mathbf{i} - 5\mathbf{j}$ and $\overrightarrow{OT} = 2\mathbf{i} + 3\mathbf{j}$.
Find vector $\overrightarrow{ST}$.



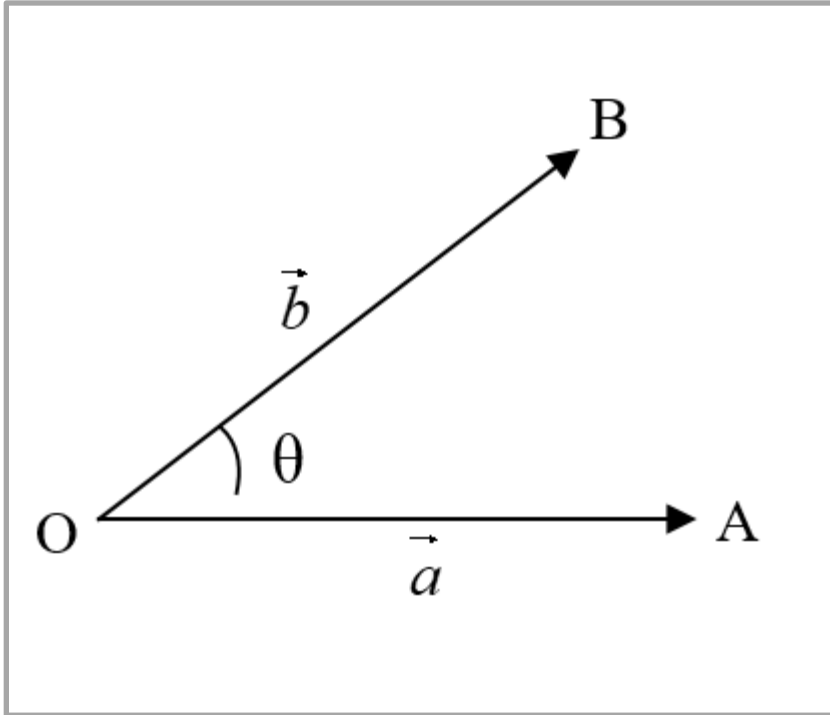
Answer:

$$7\mathbf{i} + 8\mathbf{j}$$

LEARNING OUTCOMES

- Express vector 
- Operations of vectors 
- Scalar (dot) product
- Vector (cross) product

SCALAR (DOT) PRODUCT



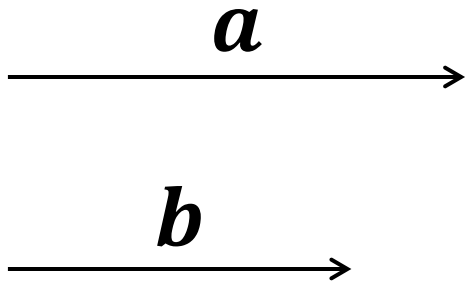
If $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and θ is the angle between $\vec{a}$ and $\vec{b}$, the scalar product is given by:

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Where $0^\circ \leq \theta \leq 180^\circ$

SCALAR (DOT) PRODUCT

➤ Parallel vector

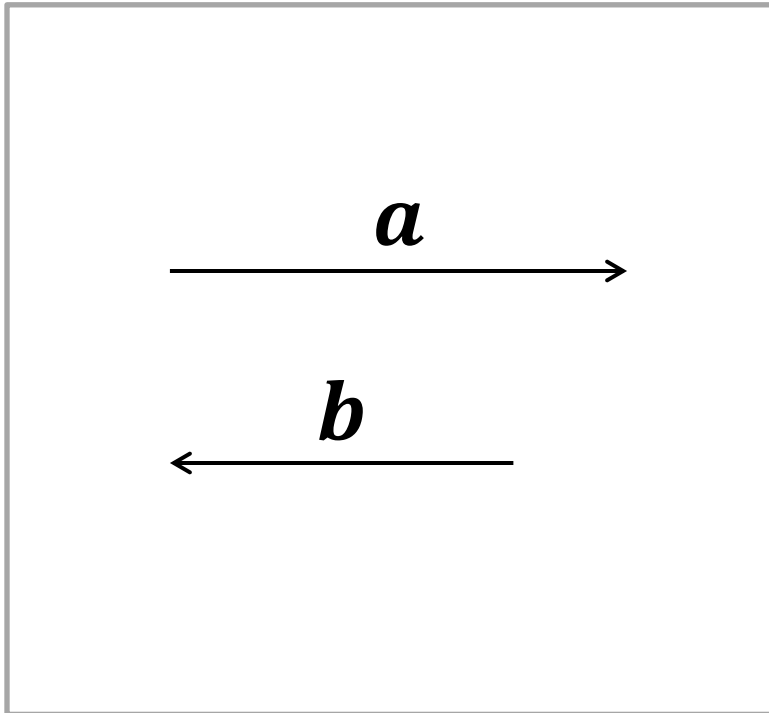


➤ θ between $\mathbf{a}$ and $\mathbf{b}$ is 0°

$$\begin{aligned}\text{➤ } \mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \\ &= |\mathbf{a}||\mathbf{b}|\cos 0^\circ \\ &= |\mathbf{a}||\mathbf{b}|(1) \\ &= |\mathbf{a}||\mathbf{b}|\end{aligned}$$

SCALAR (DOT) PRODUCT

➤ Parallel vector

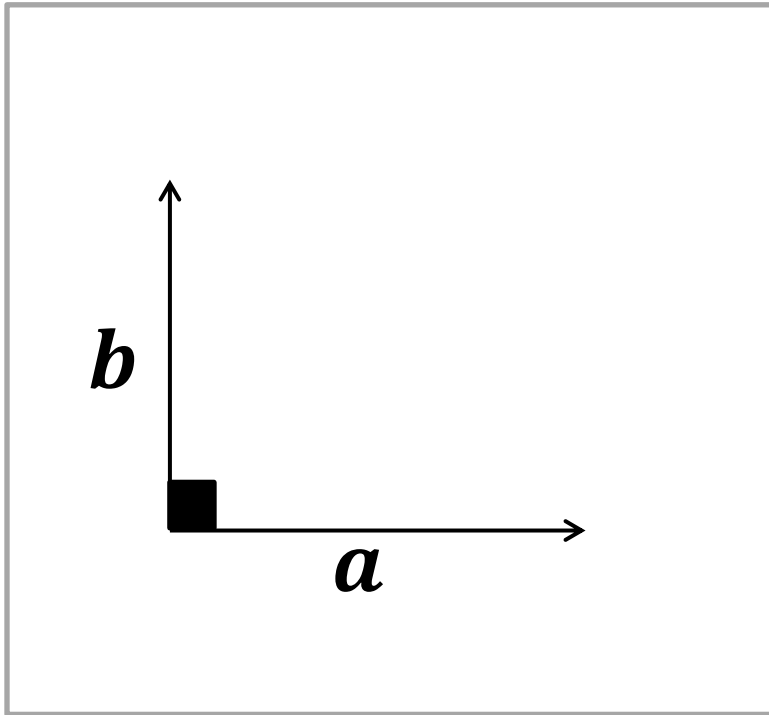


➤ θ between $\mathbf{a}$ and $\mathbf{b}$ is 180°

$$\begin{aligned}\text{➤ } \mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \\ &= |\mathbf{a}||\mathbf{b}|\cos 180^\circ \\ &= |\mathbf{a}||\mathbf{b}|(-1) \\ &= -|\mathbf{a}||\mathbf{b}|\end{aligned}$$

SCALAR (DOT) PRODUCT

➤ Perpendicular vector



➤ θ between a and b is 90°

$$\begin{aligned}\text{➤ } a \cdot b &= |a||b|\cos\theta \\ &= |a||b|\cos 90^\circ \\ &= |a||b|(0) \\ &= 0\end{aligned}$$

SCALAR (DOT) PRODUCT

➤ Properties of Scalar Product

$$\textcircled{1} \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$

$$\textcircled{2} (\omega \mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\omega \mathbf{b}) = \omega(\mathbf{a} \cdot \mathbf{b})$$

$$\textcircled{3} \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$$

$$\textcircled{4} (\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$$

$$\textcircled{5} (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{c} + \mathbf{d}) = \mathbf{a} \cdot \mathbf{c} + \mathbf{a} \cdot \mathbf{d} + \mathbf{b} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{d}$$

$$\textcircled{6} \mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$$

SCALAR (DOT) PRODUCT

➤ Result of Scalar Product

Let $\vec{A} = a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$ and
 $\vec{B} = a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}$

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}) \cdot (a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}) \\ &= a_1a_2 + b_1b_2 + c_1c_2\end{aligned}$$

SCALAR (DOT) PRODUCT

- Angle between two vectors

From $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

EXERCISE 5

① Given that $\mathbf{a} = 4\mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$ and

$\mathbf{b} = \mathbf{j} - 5\mathbf{k}$, find:

a) $\mathbf{a} \cdot \mathbf{b}$

b) angle between these two vectors

EXERCISE 5

① Answer:

a) Given: $\mathbf{a} = 4\mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$
 $\mathbf{b} = \mathbf{j} - 5\mathbf{k}$

$$\mathbf{a} \cdot \mathbf{b} = a_1 a_2 + b_1 b_2 + c_1 c_2$$

$$= 4(0) + (-2)(1) + (-4)(-5)$$

$$= 18$$

EXERCISE 5

① Answer:

$$\text{b) } \mathbf{a} = 4\mathbf{i} - 2\mathbf{j} - 4\mathbf{k} \quad , \quad \mathbf{b} = \mathbf{j} - 5\mathbf{k}$$

$$\mathbf{a} \cdot \mathbf{b} = 18$$

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

❖ Need to find $|\mathbf{a}|$ and $|\mathbf{b}|$ first

EXERCISE 5

① Answer:

$$\text{b) } \mathbf{a} = 4\mathbf{i} - 2\mathbf{j} - 4\mathbf{k} \quad , \quad \mathbf{b} = \mathbf{j} - 5\mathbf{k}$$

$$\begin{aligned} |\mathbf{a}| &= \sqrt{4^2 + (-2)^2 + (-4)^2} \\ &= \sqrt{36} = 6 \end{aligned}$$

$$\begin{aligned} |\mathbf{b}| &= \sqrt{0^2 + 1^2 + (-5)^2} \\ &= \sqrt{26} \end{aligned}$$

EXERCISE 5

① Answer:

$$\text{b) } |\mathbf{a}| = 6 \quad , \quad |\mathbf{b}| = \sqrt{26} \quad , \quad \mathbf{a} \cdot \mathbf{b} = 18 \quad \checkmark$$

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$$

$$= \frac{18}{6\sqrt{26}}$$

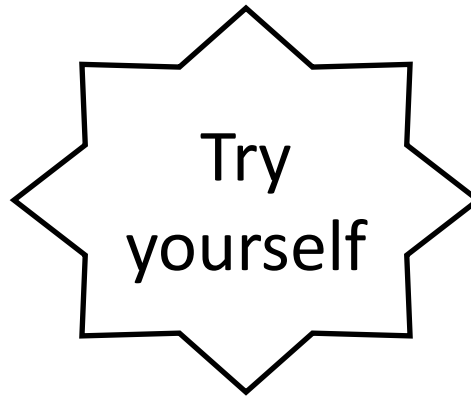
$$= 0.588 \quad \checkmark \quad \checkmark$$

$$\theta = \cos^{-1} 0.588$$

$$= 53.98^\circ$$

EXERCISE 5

- ② Find the angle between vectors
 $\overrightarrow{OP} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{OQ} = \mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$.

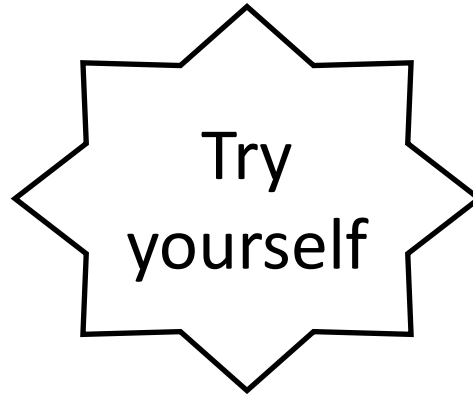


Answer:

116.87°

EXERCISE 5

- ③ Determine the angle between vectors $\overrightarrow{KL}$ and $\overrightarrow{KM}$ if $K(5, -3, 2)$, $L(0, 0, 2)$ and $M(3, 6, -2)$



Answer:

50.88°

LEARNING OUTCOMES

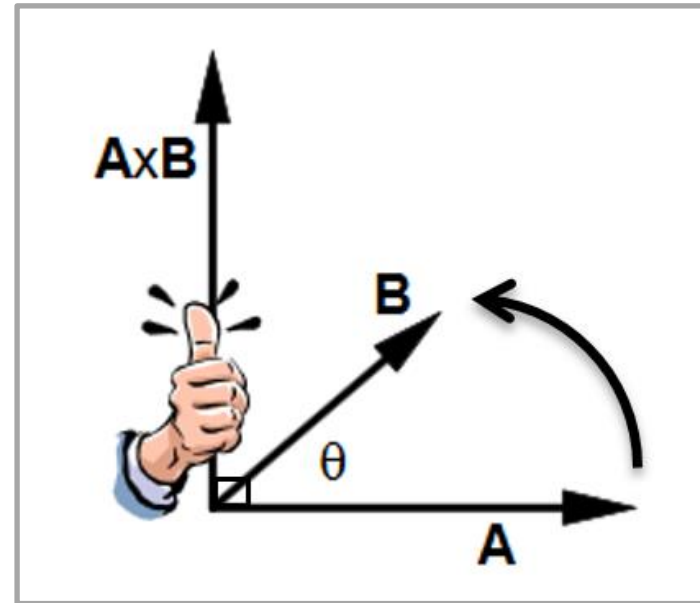
- Express vector 
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- Vector (cross) product

VECTOR (CROSS) PRODUCT

Let $\overrightarrow{OA} = a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$ and

$$\overrightarrow{OB} = a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}$$

$$\overrightarrow{OA} \times \overrightarrow{OB} = \begin{vmatrix} \overset{+}{\mathbf{i}} & \overset{-}{\mathbf{j}} & \overset{+}{\mathbf{k}} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$



$$= (b_1c_2 - c_1b_2)\mathbf{i} - (a_1c_2 - c_1a_2)\mathbf{j} + (a_1b_2 - b_1a_2)\mathbf{k}$$

VECTOR (CROSS) PRODUCT

➤ Properties of Vector Product

$$\textcircled{1} \mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$$

$$\textcircled{2} \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$$

$$\textcircled{3} \mathbf{a} \times \mathbf{a} = \mathbf{0}$$

$$\textcircled{4} \omega(\mathbf{a} \times \mathbf{b}) = (\omega\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (\omega\mathbf{b})$$

$$\textcircled{5} \text{ If } \mathbf{a} \times \mathbf{b} = \mathbf{0}, \text{ then } \mathbf{a} \text{ and } \mathbf{b} \text{ are parallel}$$

EXERCISE 6

① If $\overrightarrow{OR} = -\mathbf{i} + 9\mathbf{j} - 5\mathbf{k}$ and $\overrightarrow{OS} = 3\mathbf{i} + 6\mathbf{k}$,

find:

a) $\overrightarrow{OR} \cdot \overrightarrow{OS}$

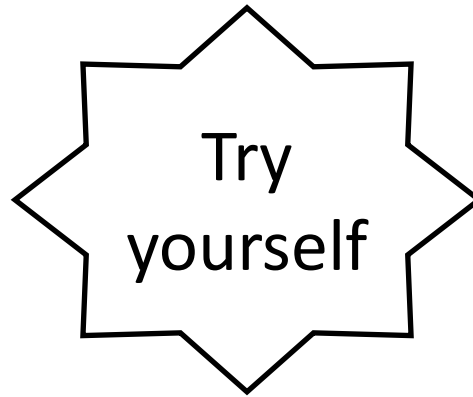
b) $\overrightarrow{OR} \times \overrightarrow{OS}$

EXERCISE 6

① Answer:

$$\text{a) } \overrightarrow{OR} = -\mathbf{i} + 9\mathbf{j} - 5\mathbf{k} \quad , \quad \overrightarrow{OS} = 3\mathbf{i} + 6\mathbf{k}$$

$$\overrightarrow{OR} \cdot \overrightarrow{OS}$$



Answer:

$$-33$$

EXERCISE 6

① Answer:

$$\text{b) } \overrightarrow{OR} = -\mathbf{i} + 9\mathbf{j} - 5\mathbf{k} \quad , \quad \overrightarrow{OS} = 3\mathbf{i} + 6\mathbf{k}$$

$$\overrightarrow{OR} \times \overrightarrow{OS} = \begin{array}{c} \textcolor{red}{+} \quad \textcolor{red}{-} \quad \textcolor{red}{+} \\ \left| \begin{array}{ccc} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 9 & -5 \\ 3 & 0 & 6 \end{array} \right| \end{array}$$

$$= [9(6) - (-5)(0)] \mathbf{i} - [(-1)(6) - (-5)(3)] \mathbf{j} + [(-1)(0) - 9(3)] \mathbf{k}$$

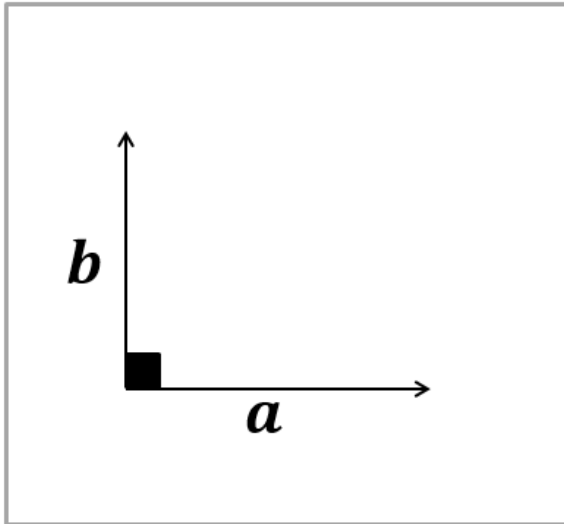
$$= 54\mathbf{i} - 9\mathbf{j} + (-27)\mathbf{k} = 54\mathbf{i} - 9\mathbf{j} - 27\mathbf{k}$$

EXERCISE 6

②

SCALAR (DOT) PRODUCT

➤ Perpendicular vector



➤ θ between a and b is 90°

$$\begin{aligned}\text{➤ } a \cdot b &= |a||b|\cos\theta \\ &= |a||b|\cos 90^\circ \\ &= |a||b|(0) \\ &= 0\end{aligned}$$

$$a \cdot b = 0$$

EXERCISE 6

② Answer:

$$\mathbf{u} = 8\mathbf{i} + 2p\mathbf{j} + 3\mathbf{k} , \quad \mathbf{v} = 10\mathbf{i} - 11\mathbf{j} - 6p\mathbf{k}$$

$$\mathbf{u} \cdot \mathbf{v} = 0$$

$$(8\mathbf{i} + 2p\mathbf{j} + 3\mathbf{k}) \cdot (10\mathbf{i} - 11\mathbf{j} - 6p\mathbf{k}) = 0$$

$$8(10) + 2p(-11) + 3(-6p) = 0$$

$$80 - 22p - 18p = 0$$

$$80 - 40p = 0$$

$$-40p = -80$$

$$p = 2$$

EXERCISE 6

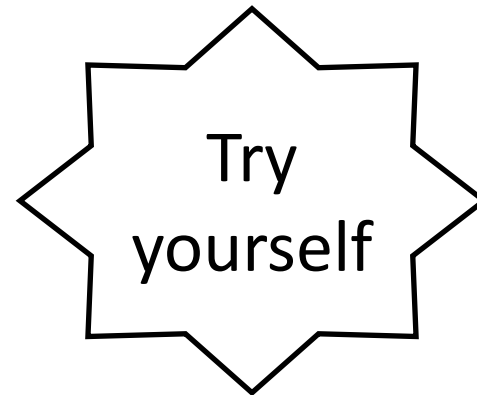
- ③ Given the vectors of $\overrightarrow{OA} = 2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$,
 $\overrightarrow{OB} = 6\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\overrightarrow{OC} = 4\mathbf{i} - \mathbf{j} + 5\mathbf{k}$.

Determine:

a) $\overrightarrow{BA} \cdot \overrightarrow{BC}$

b) $\overrightarrow{AB} \times \overrightarrow{BC}$

c) $|\overrightarrow{AB} \times \overrightarrow{BC}|$



EXERCISE 6

③ Answer:

a) 23

b) $-15\mathbf{i} - 18\mathbf{j} - 14\mathbf{k}$

c) 27.295

EXERCISE 6

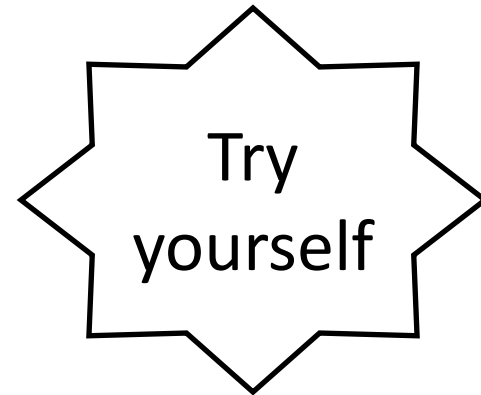
④ If $\overrightarrow{OR} = 5\mathbf{i} - 2\mathbf{k}$, $\overrightarrow{OS} = \mathbf{i} + \mathbf{j} - 4\mathbf{k}$ and $\overrightarrow{OT} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$. Find:

a) $\overrightarrow{OR} \cdot (\overrightarrow{OS} \times \overrightarrow{OT})$

b) $\overrightarrow{OT} \cdot (\overrightarrow{OR} \times \overrightarrow{OS})$

c) $\overrightarrow{OR} \times (\overrightarrow{OS} \times \overrightarrow{OT})$

d) $(\overrightarrow{OR} \times \overrightarrow{OS}) \times \overrightarrow{OT}$



EXERCISE 6

④ Answer:

a) 7

b) 7

c) $-10\mathbf{i} + 3\mathbf{j} - 25\mathbf{k}$

d) $-59\mathbf{i} + 16\mathbf{j} - 34\mathbf{k}$

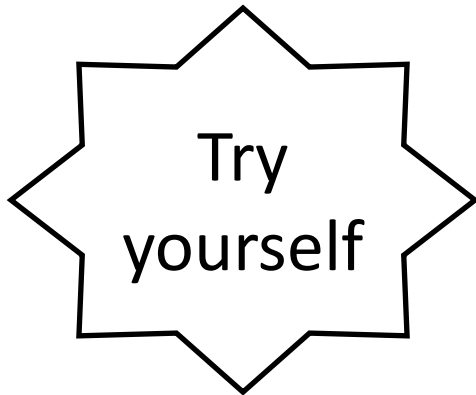
EXERCISE 6

⑤ The vector $\overrightarrow{OK} = -5\mathbf{i} + a\mathbf{j} - 2\mathbf{k}$,

$$\overrightarrow{OL} = \mathbf{i} - 2\mathbf{j} + \mathbf{k} \text{ and } \overrightarrow{OM} = b\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}.$$

Find the values of a and b if

$$\overrightarrow{KL} \times \overrightarrow{LM} = -21\mathbf{i} - 6\mathbf{j} + 36\mathbf{k}$$



Answer:

$$a = 1$$

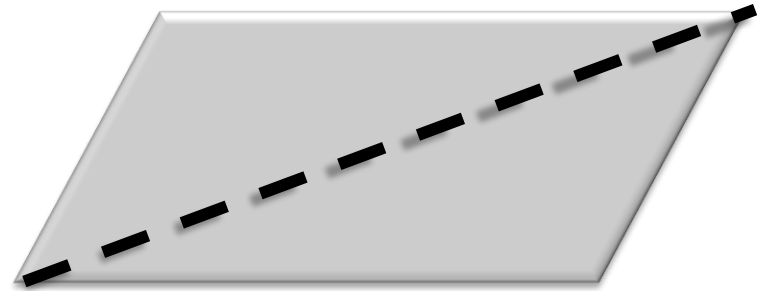
$$b = 3$$

VECTOR (CROSS) PRODUCT

- Area of triangle and parallelogram

Area of parallelogram

$$= |\vec{AB} \times \vec{BC}|$$



Area of triangle

$$= \frac{1}{2} |\vec{AB} \times \vec{BC}|$$

EXERCISE 7

- ① Find the area of triangle having vertices at $P(1,2,5)$, $Q(-2,1,2)$ and $R(-3,4,-1)$.

Answer:

Given:

$$\mathbf{P} = (1,2,5) \longrightarrow \overrightarrow{OP}$$

$$\mathbf{Q} = (-2,1,2) \longrightarrow \overrightarrow{OQ}$$

$$\mathbf{R} = (-3,4,-1) \longrightarrow \overrightarrow{OR}$$

EXERCISE 7

① Answer:

$$\text{Area of triangle} = \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{QR}|$$

1

$\overrightarrow{PQ}$ AND $\overrightarrow{QR}$

2

$\overrightarrow{PQ} \times \overrightarrow{QR}$

3

$|\overrightarrow{PQ} \times \overrightarrow{QR}|$

4

$\frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{QR}|$

EXERCISE 7

① Answer:

$$\overrightarrow{PQ} = (1, -1, 3) \qquad \overrightarrow{QR} = (1, -2, -1)$$

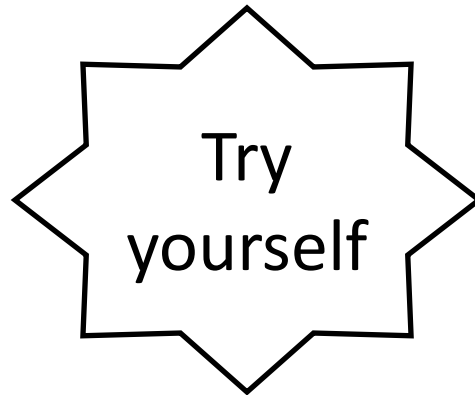
$$\overrightarrow{PQ} \times \overrightarrow{QR} = (7, 4, -1)$$

$$|\overrightarrow{PQ} \times \overrightarrow{QR}| = \sqrt{66}$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{QR}| \\ &= 4.062 \text{unit}^2 \end{aligned}$$

EXERCISE 7

- ② Find the area of parallelogram with points $K(1,2,3)$, $L(1,3,6)$ and $M(3,8,6)$.

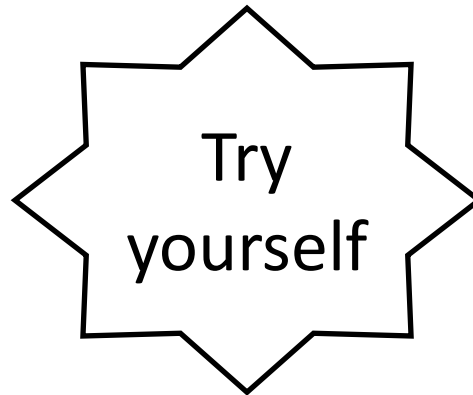


Answer:

16.279unit²

EXERCISE 7

- ③ A plane contains points $P(1, -1, 3)$, $Q(3, 1, 4)$ and $R(-2, 4, 3)$.
Find the area of triangle PQR.



Answer:

8.515unit²

LEARNING OUTCOMES

- Express vector



- Operations of vectors



- Scalar (dot) product



- Vector (cross) product



DRILLS: June 2019 Session

a) Given that $\mathbf{m} = \begin{pmatrix} -3 \\ g \\ 4 \end{pmatrix}$, $\mathbf{n} = \begin{pmatrix} -1 \\ -1 \\ h \end{pmatrix}$ and

$$3\mathbf{n} - \mathbf{m} = \begin{pmatrix} 0 \\ -5 \\ 8 \end{pmatrix}.$$

i) Find the value of g and h . [3 marks]

ii) Calculate $-\mathbf{n} + \mathbf{m}$. [2 marks]

iii) Calculate $3\mathbf{m} - \mathbf{n}$. [2 marks]

DRILLS: June 2019 Session

- b) i) Given that $P(-3,4)$. Calculate the unit vector in the direction of $\overrightarrow{OP}$. [4 marks]
- ii) Given that vectors $\overrightarrow{OP} = -\mathbf{i} - 5\mathbf{j} - 11\mathbf{k}$ and $\overrightarrow{OR} = -4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$. Express the vectors of $\overrightarrow{PR}$ and $\overrightarrow{OR} - \overrightarrow{PO}$. [4 marks]

DRILLS: June 2019 Session

c) A triangle with vertices $P(2,0,-2)$, $Q(5,-2,3)$ and $R(0,5,4)$. Calculate:

i) $\overrightarrow{PQ} \cdot \overrightarrow{QR}$ [5 marks]

ii) $\overrightarrow{PQ} \times \overrightarrow{QR}$ [3 marks]

iii) Area of parallelogram based on the answer
on (ii) [2 marks]

DRILLS: December 2018 Session

a) Given that the position vectors of point A , B and C with respect to the origin are

$$\overrightarrow{OA} = 3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}, \overrightarrow{OB} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k} \text{ and}$$

$$\overrightarrow{OC} = -\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}. \text{ Compute:}$$

i) $\overrightarrow{AB}$ [3 marks]

ii) $\overrightarrow{BC}$ [3 marks]

iii) $\overrightarrow{AB} \cdot \overrightarrow{BC}$ [4 marks]

DRILLS: December 2018 Session

b) Given vector $\vec{A} = 6\mathbf{i} - 14\mathbf{j} + 4\mathbf{k}$ and $\vec{B} = 20\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}$. Calculate:

i) $\vec{A} - \vec{B}$ [3 marks]

ii) $\vec{A} \times \vec{B}$ [3 marks]

iii) $|\vec{AB}|$ [5 marks]

iv) Prove that the vector A is perpendicular to vector B. [4 marks]